

Establishing credibility of implicit finite element models for hip resurfacing load transfer and stress shielding: a two-model, two-mechanism V&V40 study

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Abstract

Purpose: To establish and communicate the credibility of two implicit finite element analysis (FEA) models for predicting load transfer and proximal stress shielding in cemented femoral hip resurfacing, we executed a two-model, two-mechanism study compliant with the intent of ASME V&V40. The study focuses on two clinically relevant mechanisms: (M1) stress shielding during level walking, and (M2) superior femoral neck tensile strain concentration under varus placement during stair descent. **Methods:** We assessed a homogeneous bonded-interface model (Model H) and a patient-specific heterogeneous model with cohesive cement–bone interaction and frictional cap–cement contact (Model P). Both models were solved with implicit, quasi-static FEA and were exercised over activity-specific load cases derived from instrumented gait/stair databases. Validation used cadaveric tests with resurfacing components (Co–Cr cap, PMMA mantle, short peg), strain-gauge and digital image correlation (DIC) measurements, and post-test cone-beam CT to verify implant seating and varus placement. Quantities of interest (QoIs) were the proximal load share index (LSI), cortical strain fields and strain energy density (SED) ratios for M1, and peak superior femoral neck principal tensile strain magnitude and location for M2. Credibility was graded using a private 0–5 Credibility Level Index (CLI) with acceptance at CLI $\geq$ 3.5, across six factors: representation choice, discretization effects, agreement with measurements, centrality of chosen outputs, clinical representativeness of tests, and fidelity of applied loads and constraints. **Results:** For M1, Model H achieved RMS cortical strain error 8.4% versus gauges and LSI difference 0.03 absolute; discretization effects changed LSI by 0.9% upon mesh halving. For M2, Model H underpredicted peak superior neck tensile strain by 22% with a broader strain band than DIC. For Model P, M1 RMS strain error was 6.1% (gauges) and 7.4% (DIC), with LSI difference 0.02; discretization altered LSI by 1.7%. For M2, Model P matched peak tensile strain within 9.6% and its location within 3.4 mm, with mesh/contact refinements changing peak strain by 3.1%. Test rigs replicated hip contact force magnitude within 5–9% and direction within 3–6°, and achieved target varus placement of $4.8 \pm 0.7^\circ$. **Conclusions:** Within the stated Context of Use (COU) for pre-clinical evaluation of resurfacing load transfer and neck safety factors under the two targeted mechanisms, Model P met or exceeded CLI $\geq$ 3.5 for all six credibility factors and both mechanisms. Model H met the same criterion for M1 but did not for M2 due to the absence of interfacial failure mechanisms and limited ability to localize neck tensile strains. The study delineates when a simplified bonded-interface model is sufficient for decisions around proximal shielding during walking and when a cohesive-interface model is warranted to evaluate superior neck safety in varus placement under stair descent loads. **Impact:** The results inform model selection for device design and for orthopaedic pre-clinical assessment and provide a transparent, mechanism-specific credibility record aligned with the V&V40 framework.

Keywords: hip resurfacing, finite element credibility, V&V40

1. Introduction

Cemented femoral hip resurfacing seeks to restore joint function while preserving bone stock by capping the femoral head with a metal component fixed through a thin polymethyl methacrylate (PMMA) mantle and a short central peg. While resurfacing has delivered good outcomes in select cohorts, early failures have included femoral neck fracture and progressive proximal bone loss in some patients. Device and surgical planning communities have turned to computational mechanics to understand and mitigate these mechanisms. However, models differ in their idealizations of bone heterogeneity and interfacial behavior, and their acceptance for decision-making requires

structured credibility assessment.

The ASME V&V40 framework provides a risk-informed approach to showing that a model is fit for a stated purpose. It calls for an explicit clinical or device decision context, a transparent relationship between validation evidence and the use case, and a set of credibility activities tailored to the model's impact on that decision. We adopted this approach to evaluate two implicit finite element models for hip resurfacing load transfer and stress shielding, emphasizing two mechanisms: chronic proximal stress shielding during level walking (M1) and tensile strain concentration in the superior femoral neck under varus component placement during stair descent (M2).

Our assessment was designed around two complementary

models. Model H is deliberately simple: homogeneous, linear-elastic bone and fully bonded interfaces, representing a minimal model that can be calibrated once and applied broadly. Model P is richer: patient-specific, CT-mapped heterogeneity with orthotropic bone behavior and interfacial mechanics through a cohesive cement–bone law and frictional cap–cement contact. Both models are solved with implicit quasi-static FEA and report quantities aligned with established mechanobiologic and fracture risk concepts. The intent of the study is not to rank models generally but to establish when each model can credibly inform specific, mechanism-driven questions.

We use a private Credibility Level Index (CLI) on a 0–5 scale to grade evidence for six factors important to stakeholders. A pass is defined as $CLI \geq 3.5$. We present the evidence and gradations separately for each model and mechanism to make clear where stronger physics is needed and where simpler approximations suffice. The paper reports, in detail, the modeling approaches, loading conditions, experimental comparators, and the mapping from data to the decision context, along with numerical details adequate for replication.

Relevance of the validation activities to the COU — gradation.

- a. Bench tests or data sources unrelated to the clinical scenario without justification
- b. At least one load magnitude or boundary configuration representative of the clinical scenario is used
- c. Multiple representative loads and constraints; geometric and implant configurations documented
- d. Activity-specific loads with uncertainty bounds; implant placement and fixation states verified post hoc
- e. Activity-specific loads with subject-specific kinematics and kinetics; imaging confirms implant seating and soft tissue conditions

2. Device background and clinical rationale

Femoral hip resurfacing replaces only the articular surface of the femoral head with a contoured metal component, typically cobalt–chromium–molybdenum (Co–Cr–Mo), fixed to the prepared head and neck with a thin PMMA cement mantle and a short central peg. The surgical objective is to retain native bone stock, optimize head–neck offset, and deliver favorable load transfer through the proximal femur. Compared to total hip arthroplasty, resurfacing preserves the femoral canal and can offer improved stability in young, active patients; however, it places greater demands on the femoral neck and on the integrity of the cement–bone interface.

Two failure mechanisms are particularly salient. First, chronic reduction of mechanical stimulus in the proximal femur may drive adaptive resorption and thinning of supportive trabeculae—proximal stress shielding—undermining support for

the cement mantle and cap. Second, malposition that shifts the cap in varus can elevate tensile strains along the superior aspect of the neck during stair descent or rising from a chair, increasing the risk of early neck fracture. Both mechanisms depend on the distribution of load between the cap–cement construct and the neck cortex, and on the micromechanics of cement–bone interaction in different regions.

From a design and pre-clinical standpoint, it is essential to quantify how a resurfacing construct redistributes load during daily activities and whether a given alignment creates unsafe tensile bands. Computational models that resolve strains, interface tractions, and energy densities at the relevant scales can provide this information before expensive or ethically constrained clinical evaluations. The credibility of those models needs to be established with appropriate evidence to support decisions such as geometry refinements, surgical guides, or acceptance of safety margins.

Relevance of the quantities of interest — gradation.

- a. Outputs are chosen for convenience with no linkage to clinical or design decisions
- b. Outputs qualitatively relate to the decision but lack quantitative mapping or thresholds
- c. Outputs correspond to known biomechanical metrics with literature support
- d. Outputs map to decision thresholds or margins supported by mechanobiologic or clinical studies
- e. Outputs are directly tied to pre-specified acceptance criteria for the decision, with sensitivity to confounders documented

3. Question of Interest (QOI)

We address two interrelated questions aligned with mechanisms M1 and M2.

For M1 (proximal stress shielding during level walking): Does the resurfacing construct reduce the proximal femoral mechanical stimulus, as reflected by cortical/cancellous strain fields and strain energy density, compared with intact loading, and by how much does the load share between proximal regions and the mid-diaphysis change under gait-level joint and muscle forces?

For M2 (femoral neck tensile strain concentration under varus placement during stair descent): When the femoral component is placed in varus (reduced neck–shaft angle relative to pre-operative), does the superior neck experience elevated tensile strains, and are these strains sufficiently localized and elevated to warrant fracture risk mitigation?

These questions are operationalized into QoIs: for M1, the proximal load share index (LSI), defined as the ratio of proximal (head–neck junction to 25 mm distal) cortical SED to mid-diaphyseal cortical SED; and, for M2, the peak first

principal tensile strain and its centroidal location along the superior neck cortex within 35 mm distal to the head–neck junction. Secondary QoIs include interface tractions and cement mantle stresses that contextualize the primary indices.

Output comparison — gradation.

- a. Qualitative comparison only; no metrics or uncertainty
- b. Single metric against a single test with no measurement uncertainty
- c. Multiple metrics with measurement uncertainty, but limited spatial coverage or sample size
- d. Spatially resolved comparisons (e.g., DIC or gauge arrays), with statistical error metrics across specimens
- e. Multi-modal comparisons (gauges, DIC, CT), predefined acceptance bands, and outlier analysis with justifications

4. Context of Use (COU) and rationale

The models are intended for pre-clinical evaluation of hip resurfacing constructs and surgical alignments, to inform decisions on design iteration and surgical guidance within two specific clinical activities: level walking at self-selected speed (M1) and stair descent at a controlled cadence (M2). Within this COU, models are used to estimate how resurfacing alters proximal load transfer relative to intact bone and to assess whether varus component placement induces unsafe tensile strain concentrations in the superior femoral neck during stair descent.

The rationale for this scope is twofold. First, level walking dominates daily loading cycles and has well-characterized joint contact and muscle force patterns from instrumented implants and musculoskeletal models, allowing representative application to cadaveric tests and simulations. Second, stair descent is a commonly reported contributor to high tensile demand along the superior neck due to abductor and vasti force vectors combined with adduction moments, particularly when the cap is in varus. By constraining the COU to these mechanisms and activities, we can credibly align the tests and models without conflating other activities that may have different dominant physics (e.g., stumble recovery).

The intended users are device engineers and research orthopaedic surgeons overseeing design choices and alignment strategies. The models are not intended for individualized clinical risk prediction in single patients under arbitrary activities; outside this COU, further evidence would be necessary.

Relevance of the quantities of interest — gradation.

- a. Outputs lack a clear pathway to influence the stated decision
- b. Outputs offer qualitative direction but do not constrain decisions

- c. Outputs provide quantitative margins that inform design or alignment choices
- d. Outputs correspond to margins traceable to repeatable measurements in validation tests
- e. Outputs are linked to both validation and clinical thresholds, with sensitivity to parameter variability quantified

5. Model risk level and rationale

We assessed model risk as the combination of decision consequence and model influence within the COU. Consequences are moderate to high: misplaced confidence in proximal load transfer could promote designs that shield bone, while misjudging neck strain concentration under varus could underestimate fracture risk. Model influence is moderate: bench tests and literature provide guardrails, and the models serve as one of several inputs to design choice rather than as the sole arbiter. The resulting risk categorization is medium-to-high, commensurate with the need for quantitative validation against experimental data and for targeted evaluation of interfacial mechanics.

The selected credibility activities, including specimen-specific validation for Model P and mechanism-targeted comparators for both models, reflect this risk categorization. Where peak tensile strain under varus is concerned, we emphasized interfacial representation and spatially resolved strain measurements. For proximal stress shielding, we emphasized energy-based QoIs and load partitioning that are less sensitive to narrow bands of tensile strain but are sensitive to bulk load transfer through the metaphyseal cortical shell and cancellous network.

Test conditions — gradation.

- a. Loads and boundary conditions not documented; unknown fidelity to the clinical situation
- b. Aggregate load magnitudes documented; directions or muscle forces missing
- c. Joint and muscle loads documented with ranges; mounting and constraints approximated
- d. Full loading vectors with uncertainties; mounting verified against anatomical axes; implant alignment measured
- e. Specimen-specific loading states reconstructed from motion analysis; boundary friction/compliance characterized and included

6. Mechanisms evaluated (M1, M2)

We defined two mechanisms of interest. M1 addresses chronic proximal stress shielding during level walking. In this mechanism, the resurfacing construct—via the stiffness of the

cap and cement and their coupling to the bone—redistributes load such that cancellous and cortical stimulation in the proximal femur may decrease relative to the intact configuration. This is manifest as reduced SED in proximal regions and altered strain gradients across the head–neck junction toward the diaphysis. Because remodeling stimuli scale with strain energy and microstrain magnitudes, a persistent reduction may elicit trabecular thinning. Capturing M1 requires accurate representation of global and regional stiffness and realistic walking loads and muscle forces, but is relatively insensitive to localized interfacial failure bands so long as the interfaces transfer load without gross slip.

M2 addresses the formation of a tensile strain concentration in the superior femoral neck under varus placement during stair descent. In this mechanism, the adduction moment and muscle force distribution, together with a cap placed in varus relative to the native neck–shaft angle, elevate tensile stress at the superior cortex. When the cement–bone interface permits local debonding or microseparation, the strain may localize into a narrow band whose amplitude and centroid increase neck fracture risk. Capturing M2 requires resolution of interfacial micromechanics, sufficient mesh density in the superior neck cortex and mantle, and a stair descent load path that is realistic for the varus-aligned construct.

Model form — gradation.

- a.
Model equations and constitutive choices are undocumented or ad hoc
- b.
Constitutive laws and boundary idealizations documented but not justified against mechanisms of interest
- c.
Model equations justified against dominant physics; known omissions listed
- d.
Model includes mechanism-relevant nonlinearities or interfaces; omissions justified as non-dominant for COU
- e.
Alternative model forms explored or bounded; chosen form shown to be sufficient for QoIs with sensitivity analyses

7. Computational models (Model H and Model P): geometry, material laws, and boundary conditions

Geometry and meshing: We generated specimen-specific femoral geometries from clinical-resolution CT (in-plane 0.7–0.9 mm, slice thickness 1.0 mm) using semi-automated segmentation, with manual refinement of the head–neck junction. The resurfacing component geometry was a 48–54 mm outer diameter Co–Cr–Mo cap featuring a 1.5 mm nominal cement mantle and a 14 mm long, 8 mm diameter central peg with 2° taper. Implant positioning followed standard resurfacing preparation with a central guidewire aligned along the femoral neck axis; for M2, the cap was intentionally placed in varus relative to the native neck–shaft angle by approximately 5°.

Meshes were tetrahedral (10-node quadratic elements) for bone and cement, with hexahedral wedges used where topology allowed in the mantle to reduce locking. Local mesh refinement to an edge length of 0.5 mm was applied across the superior neck cortex, mantle, and interface zones; coarser elements (1.5–2.0 mm) were used in the distal diaphysis.

Material laws: In Model H, cortical and cancellous bone were represented as a single homogeneous linear-elastic material ($E = 12.5$ GPa, $\nu = 0.3$) to represent an effective stiffness for the proximal femur. The Co–Cr–Mo cap was modeled as linear-elastic isotropic ($E = 210$ GPa, $\nu = 0.30$), and PMMA cement as linear-elastic isotropic ($E = 2.4$ GPa, $\nu = 0.35$). The cement–bone and cap–cement interfaces were fully bonded with shared nodes, representing perfect adhesion without slip.

In Model P, bone orthotropy and heterogeneity were mapped from CT Hounsfield Units (HU) to tissue apparent density using a bilinear calibration from a phantom, and then to orthotropic elastic constants. For cortical regions (thresholded by density and morphology), the longitudinal modulus E_x was defined along the principal fabric direction from the structure tensor, with transverse moduli E_y and E_z reduced per standard fabric–elasticity relations (e.g., $E_x = 14.5\rho_{app}^{1.8}$ GPa, $E_y = E_z = 0.55E_x$). For trabecular regions, moduli followed $E_x = 9.0\rho_{app}^{1.8}$ GPa, $E_y = E_z = 0.60E_x$, with Poisson ratios $\nu_{xy} = \nu_{xz} = 0.28$ and $\nu_{yz} = 0.24$. The cap and cement laws were as in Model H.

Interfaces in Model P: The cement–bone interface was represented with zero-thickness cohesive elements using a bilinear traction–separation law with normal and shear strengths of 2.5 and 3.5 MPa, respectively, and a fracture energy of 0.30 N/mm (mode I) and 0.50 N/mm (mode II). Damage initiation followed a quadratic nominal stress criterion, and evolution followed an energy dissipation law with linear softening. Cohesive elements were seeded over the entire mantle–bone contact area with an initial stiffness sufficient to prevent artificial compliance ($K_n = K_s = 10^6$ MPa/mm). The cap–cement interface used frictional contact with a penalty formulation and a coefficient of friction $\mu = 0.3$, with a 0.1 mm tied region at the peg–cement to represent interdigitation around the peg flutes.

Boundary conditions and loads: For both models, the distal femur was constrained with a distributed coupling over a 30 mm long cylindrical segment, permitting physiologic transverse deformation while preventing rigid-body motion. Loads were applied as joint reaction and muscle forces consistent with level walking (M1) and stair descent (M2) at the mid-stance and early-stance peaks, respectively. Hip joint contact forces followed the proportions and directions from published instrumented implant datasets, scaled to a body mass of 75–85 kg (peak magnitudes: M1, $2.7\text{--}3.2 \times$ body weight; M2, $3.4\text{--}3.8 \times$ body weight). Muscle forces included abductor, vasti, rectus femoris, hamstrings, and iliopsoas vectors applied at anatomical attachment areas as distributed traction fields, with magnitudes derived from inverse dynamics solutions tuned to match the target joint reaction within 3–6%. Force directions were aligned to anatomical axes with an uncertainty band of $\pm 3\text{--}6^\circ$. Loading ramped in ten equal steps to the peak, with quasi-static equilibrium solved at each step.



Figure 1: Schematic of Model P showing CT-mapped orthotropy in the femoral head and neck, a thin PMMA mantle with cohesive cement–bone interface (purple interface layers), and frictional cap–cement contact. Local mesh refinement (0.5 mm) is shown along the superior femoral neck.

Model form — gradation.

- a. Boundary and interface idealizations are selected for computational convenience
- b. Elastic properties and interface choices are referenced but not connected to load transfer mechanisms
- c. Elastic and interfacial models are selected to capture global and local load sharing; interfacial damage omitted where deemed secondary
- d. Elastic heterogeneity and cohesive interactions are included to represent local debonding and frictional slip under load
- e. Parallel model forms (e.g., bonded vs cohesive) are exercised to bound QoIs and assess sensitivity to interface physics

8. Quantities of interest and their relevance

Primary QoIs were chosen to map directly onto the two mechanisms. For M1, we computed the proximal load share index (LSI) as the ratio of volume-averaged cortical SED in a proximal region (from the head–neck junction to 25 mm distal, excluding cement and cap) to the volume-averaged cortical SED in a mid-diaphyseal region (a 20 mm long segment centered 120 mm distal to the lesser trochanter). When compared to intact configuration values, reductions in LSI indicate proximal shielding. We also extracted cortical and cancellous strain distributions in the proximal region to contextualize energy measures and to visualize redistribution across the head–neck junction.

For M2, we computed the peak first principal tensile strain, ϵ_1 , along the superior femoral neck cortex within 35 mm distal to the head–neck junction, and recorded its centroidal location along the neck axis. We compared these fields and peak values to experimental DIC and gauge measurements to evaluate localization and amplitude of the tensile band. Secondary QoIs for M2 included cement mantle maximum principal tensile stress and interface traction fields, which are mechanistically

informative for where strain localizes under different interfacial behaviors.

The choice of these QoIs is grounded in mechanobiology and fracture risk: energy-based measures are predictive of remodeling responses, while tensile strain magnitude and localization track with crack initiation risk in cortical bone. By constraining the QoIs to these indices, the models directly inform whether a given construct and alignment preserve proximal stimulus during walking and whether varus placement risks superior neck failure during stair descent.

Relevance of the quantities of interest — gradation.

- a. QoIs are descriptive but not connected to thresholds relevant to bone maintenance or fracture
- b. QoIs link to general mechanobiologic principles but lack operational definitions
- c. QoIs have operational definitions and correspond to published thresholds or correlates
- d. QoIs are normalized against intact baselines and support margins for design or alignment
- e. QoIs are mapped to acceptance bands used in validation, closing the loop from mechanism to decision

9. Test conditions characterization for validation and COU loads

We replicated level walking and stair descent loading states on cadaveric femora prepared with the study resurfacing component. For walking (M1), we imposed a mid-stance joint contact vector derived from instrumented total hips and musculoskeletal models, with a target magnitude of $2.9 \times$ body weight (BW) for a 80 kg surrogate and a direction $20\text{--}25^\circ$ medial-lateral from the femoral shaft axis and $15\text{--}20^\circ$ anterior-posterior relative to the frontal plane. Muscle surrogates were applied via cable-and-pulley fixtures attached at anatomical footprints with load cells to monitor applied tension. Abductor forces targeted $1.1 \times$ BW, vasti $0.4 \times$ BW, and hamstrings $0.2 \times$ BW at mid-stance to reconstruct the joint reaction and bending pattern.

For stair descent (M2), we applied an early-stance loading state with a joint reaction target of $3.6 \times \text{BW}$, directed more medially relative to the shaft axis ($25\text{--}30^\circ$) and slightly posteriorly ($5\text{--}10^\circ$) to replicate the knee extensor demand. Muscle surrogates increased abductor and vasti components to $1.3 \times \text{BW}$ and $0.6 \times \text{BW}$, respectively. The femoral component was seated with intentional varus relative to the native neck by guiding the central pin to a varus offset, yielding a final varus placement of $4\text{--}6^\circ$ as confirmed by post-test cone-beam CT.

Boundary conditions followed established potting and constraint practices: the distal 60 mm of the femur was embedded in PMMA within an aluminum pot that coupled to the test frame via a spherical bearing, constraining translations while permitting small rotations. The cap was articulated against a polyethylene cup to distribute the joint reaction vector. Preconditioning cycles (5–10 load cycles) were applied to stabilize the cement mantle and soft test fixtures. Load magnitudes and directions were recorded via a six-axis load cell beneath the cup, and line-of-action angles were measured with a digital inclinometer on a rigid arm aligned to the known femoral axes.

Across all tests, the joint contact magnitude matched targets within 5–9% (walking mean absolute error 5.8%; stair descent 6.7%). The line-of-action direction matched targets within $3\text{--}6^\circ$ (walking 3.4° ; stair descent 5.2°). Muscle surrogates were controlled to within 7–11% of desired magnitudes as measured at the load cells. These statistics define the fidelity of the applied loading states relative to the modeled loads.

Test conditions — gradation.

- a. Generic loads are used with no attempt to reproduce the COU
- b. Approximate reproductions of load magnitudes; directions and co-contractions not controlled
- c. Load magnitudes and directions reproduced within broad tolerances; at least one surrogate muscle group applied
- d. Full vector reproduction with multiple muscle surrogates; alignment and constraints verified post hoc
- e. Specimen-specific load calibration ensuring vector and moment balance; soft tissue and friction effects quantified

10. Model form justification with respect to M1 and M2

For M1, the dominant physics involve global stiffness coupling between the cap–cement construct and the proximal femur, distributing load between metaphyseal cortex and cancellous bone under mid-stance loads. In this setting, a bonded-interface idealization maintains load continuity across the mantle and into the bone and can reasonably represent the net load transfer if interfacial damage is modest. A homogeneous elastic representation of bone, while simplifying local heterogeneity, can still capture first-order redistribution of

proximal to diaphyseal energy. We therefore expected Model H to be adequate for LSI and proximal strain trends in walking.

For M2, local separation and slip at the cement–bone interface can focus tensile loads into the superior neck cortex under varus placement and stair descent loading. If the interface is assumed fully bonded, the model cannot reflect such localized load paths and may smear the tensile band. Capturing this mechanism requires interfacial traction–separation behavior and contact at the cap–cement interface to permit minor repositioning of the load path as the mantle deforms. The orthotropic heterogeneity in the neck further influences where the tensile band localizes. Model P embeds these capabilities and was expected to more realistically represent M2 outcomes.

We note that neither model includes time-dependent creep of cement, viscoelastic behavior of bone, or fatigue damage accumulation. For the quasi-static load cases and for the chosen QoIs, these omissions are not expected to strongly influence peak strain and energy distributions, though they could be relevant for long-term remodeling or fatigue assessments beyond the COU.

Model form — gradation.

- a. Single, simplified constitutive form assumed across all tissues and interfaces
- b. Different tissue laws used but interfaces constrained to perfect bond
- c. Interfaces modeled with contact or cohesive behavior, tuned from literature
- d. Interfaces and material anisotropy mapped from imaging; calibration performed to specimen characteristics
- e. Multiple interface laws bracketed; parameter uncertainty quantified and shown to leave QoI decisions invariant

$$\text{CI}_{\{\text{score}\}} = \frac{\sum_{i=1}^n w_i x_i}{\sum_{i=1}^n w_i} \quad (1)$$

$$\backslash \text{varepsilon}_{\{\text{rel}\}} = \frac{\sum_{i=1}^n w_i x_i}{\sum_{i=1}^n w_i} \quad (2)$$

$$\text{U}_{\{\text{val}\}} = \frac{\sum_{i=1}^n w_i x_i}{\sum_{i=1}^n w_i} \quad (3)$$

11. Numerical solver error estimation and mesh convergence

We used implicit quasi-static finite element solvers with Newton–Raphson iterations, line search, and adaptive load stepping. For both models, we performed h-refinement studies, solver tolerance sweeps, and contact/cohesive penalty parameter checks to quantify discretization and iterative effects on QoIs. Convergence criteria were set to 106 relative energy norm for global equilibrium and 104 relative displacement increment. Contact enforcement used a stabilized penalty method with a

target relative penetration $< 0.1\%$ of element size in Model P, and shared nodes in Model H due to bonded interfaces.

Mesh convergence was assessed by halving the element edge lengths in regions of interest (superior neck cortex, cement mantle, interfaces) and recomputing the load cases. For Model H in M1, halving the local mesh (from 1.0 to 0.5 mm near the neck; 2.0 to 1.0 mm diaphysis) changed the proximal LSI by 0.9% and peak superior neck ϵ_1 by 1.2%. In M2, the same refinement changed LSI by 1.1% and peak ϵ_1 by 1.6%. For Model P in M1, h-refinement (0.8 to 0.5 mm near interfaces; 1.5 to 1.0 mm elsewhere) changed LSI by 1.7% and peak ϵ_1 by 1.9%. In M2, with cohesive and contact active, refinement changed peak ϵ_1 by 3.1% and the location of the tensile band centroid by 2.7 mm. Solver tolerance sweeps (tightening the energy norm by one order of magnitude) altered QoIs by less than 0.4% in all cases.

Cohesive and contact penalties were checked by varying normal and tangential penalty stiffness by $\pm 50\%$ around the nominal values. In Model P M2, the variation changed peak ϵ_1 by 2.3% and the band width (full width at half maximum) by 0.9 mm, with no change in the qualitative location. These results suggest that the QoIs have limited sensitivity to the penalty choices within standard ranges. Residual norms and contact status distributions were monitored to ensure stable convergence without chattering; no step reductions below 10% of nominal increments were needed outside of the onset of cohesive softening in M2.

Numerical solver error — gradation.

- a.
No assessment of discretization or solver tolerance effects
- b.
Single mesh reported; solver tolerances set by defaults
- c.
Mesh refinement performed in regions of interest; solver tolerances reported
- d.
Mesh and solver sensitivity studies demonstrate small impact on QoIs; contact/cohesive parameters checked
- e.
Multiple discretizations and solvers compared; error estimators used to bound QoIs and propagate to decisions

12. Experimental methods for output comparison

Specimens and preparation: Fourteen fresh-frozen human femora (7 pairs; age 48–66 years; 8 male) with no prior pathology were screened radiographically. Eight were assigned to M1 walking tests and six to M2 stair descent with varus placement. Bone mineral distribution was characterized with QCT for mapping to Model P and to enable post hoc regional density analyses. Resurfacing components (Co–Cr–Mo, 48–54 mm OD) were cemented with standard PMMA under third-generation technique (vacuum mixing, retrograde injection, pressurization) by a fellowship-trained surgeon. The central peg was seated with line-to-line fit and rotational alignment along the preoperative neck axis unless modified for varus in M2.

Instrumentation: Strain measurement used a combination of uniaxial and rosette strain gauges bonded to the superior and inferior neck cortices at 10, 20, and 30 mm distal to the head–neck junction (three rosettes superiorly, three uniaxial inferiorly). Digital image correlation (DIC) provided 3D full-field surface strains over a paint-speckled superior neck region (field of view approximately 50×35 mm, facet size 19 pixels, step 6 pixels). Gauge calibration was performed via three-point bending of a surrogate beam; DIC calibration used a certified calibration grid. Sampling was at 100 Hz with peak loads held for 3 s to allow averaging over steady-state readings.

Measurement uncertainties: Gauge repeatability was ± 5 microstrain ($\mu\epsilon$) for uniaxial and ± 8 $\mu\epsilon$ for resolved rosette principal strains over three repeats. DIC strain noise floor, estimated from zero-load frames, was ± 75 $\mu\epsilon$, and spatial bias due to out-of-plane motion was mitigated with stereo camera alignment to within 0.03 mm 3D residuals. The joint contact load cell had a calibration uncertainty of $\pm 1\%$ full scale; muscle surrogate load cells $\pm 2\%$. After testing, cone-beam CT (0.5 mm voxel) verified cement mantle thickness (mean 1.41 ± 0.22 mm), peg seating, and varus angle (for M2) relative to the preoperative neck axis.

Data reduction: For comparison to models, gauge and DIC data were mapped to the femoral surface reconstructed from CT, registering optical and anatomic landmarks. Strain directions were rotated into the local principal system for ϵ_1 comparison. The LSI was derived from the intact tests (pre-resurfacing) and from the resurfaced tests by integrating SED inferred from the surface principal strains under the plane stress approximation and a cortical thickness model validated against QCT. For M2, the peak ϵ_1 and its centroid along the superior neck were extracted from DIC and registered to the model coordinate system for spatial comparison.

Output comparison — gradation.

- a.
Sparse measurements (e.g., one gauge) and qualitative agreement only
- b.
Moderate number of gauges; single load case; limited mapping to model coordinates
- c.
Gauge arrays and DIC used; mapping with documented uncertainty; single activity
- d.
Multiple activities; specimens registered to models; statistical agreement metrics reported
- e.
Comprehensive spatial comparisons with uncertainty propagation; acceptance bands defined a priori

13. Output comparison: metrics, targets, and acceptance bands

We used predefined metrics and acceptance bands to assess how closely model outputs matched measured quantities. For M1, the primary metrics were: (a) RMS error between modeled

and measured cortical principal strains at the gauge/DIC locations, normalized by the peak measured strain; an acceptance band of $\leq 15\%$ normalized RMS was set based on literature variability and measurement noise; and (b) absolute difference in LSI between model and experiment, with an acceptance band of ≤ 0.05 . Secondary metrics included correlation coefficients (target $r \geq 0.80$) between spatial strain patterns and the measured field.

For M2, the primary metrics were: (a) relative error in peak superior neck ϵ_1 (acceptance band $\leq 15\%$); (b) absolute distance between the modeled and measured centroid of the superior neck tensile band (acceptance band ≤ 5 mm); and (c) normalized RMS error across the superior neck DIC field (acceptance band $\leq 20\%$). These bands were informed by the strain noise floor, alignment mapping uncertainty, and reported inter-specimen variability of strain magnitudes under similar loads.

For both mechanisms, we also checked whether model predictions fell within the 95% confidence intervals of the experimental distributions across specimens. All comparisons accounted for measurement uncertainties by bootstrapping across measurements where applicable and by propagating load reproduction uncertainties into model predictions via ± 1 SD perturbations of joint reaction magnitude and direction.

Output comparison — gradation.

- a. No acceptance criteria; post hoc narrative fit claims
- b. Acceptance criteria stated after seeing data; few metrics
- c. Predefined metrics with literature-based bands; limited uncertainty treatment
- d. Predefined metrics and bands; uncertainty propagated; multi-specimen statistics used
- e. Decision rules pre-registered; blind validation where feasible; outlier handling specified

14. Relevance of validation activities to the COU

The validation activities were constructed to emulate the mechanical milieu of the two clinical activities within the COU. Level walking loads and muscle surrogates reconstructed a mid-stance vector proportion and direction typical of instrumented hip datasets, and the distal constraint mimicked the support provided by the knee and surrounding soft tissues as far as feasible in a benchtop test. Stair descent loads emphasized the knee extensor demand and abductor co-contraction that drive superior neck tension, with varus placement intentionally introduced and quantified via imaging. The resurfacing components and cementing technique matched contemporary surgical practice, and post-test imaging confirmed mantle thickness and seating consistent with in vivo expectations.

We further constrained the activities to mid-stance and early-stance peaks, which are the loading states with largest contributions to the QoIs in question. While time history could

be relevant for fatigue or creep, the quasi-static snapshots at peak loads were sufficient for the current questions. The tests therefore provided conditions under which the models could be evaluated against the same types of loads they would be used to analyze in the COU.

Relevance of the validation activities to the COU — gradation.

- a. Validation unrelated to the intended activity or alignment scenarios
- b. Validation partially related; some loads or alignments missing
- c. Validation covers activity peaks; implant seating and alignment documented
- d. Validation covers activity peaks with specimen-level imaging and load reconstructions
- e. Validation reproduces the full range of COU scenarios, including alignment variations and uncertainty ranges

15. Use error considerations in setup and interpretation

Potential user missteps include misregistration of specimen coordinates when mapping DIC to the model surface, misinterpretation of LSI as a surrogate for remodeling without considering local density variations, and overgeneralization of M2 findings to other activities or to valgus placement. In modeling, improper assignment of muscle force directions or magnitudes could distort the joint reaction and thereby the strain distribution. While our workflows include documented templates and checks, these considerations remain relevant to any translation or reuse of the models beyond the present study.

Use error — gradation.

- a. No consideration of user workflow leading to misuse of the model or data
- b. Documentation addresses basic inputs and outputs
- c. User guidance includes common pitfalls and checks for load and alignment setup
- d. Workflow includes automated validation of inputs (e.g., load vector balance, coordinate frames)
- e. Training, software prompts, and decision support guard against misinterpretation of QoIs within the COU

16. Credibility scoring framework (CLI 0–5) and thresholds

We scored credibility with a private Credibility Level Index (CLI) on a 0–5 scale in 0.1-unit increments to reflect gradations within and across mechanisms and models. A CLI of 0 indicates

no evidence; 1 indicates anecdotal or qualitative evidence; 2 indicates limited quantitative evidence; 3 indicates moderate quantitative evidence with gaps; 4 indicates strong quantitative evidence across key elements with minor gaps; and 5 indicates comprehensive evidence, including bracketing of model forms and robust uncertainty handling. Acceptance for use within the COU was set at $CLI \geq 3.5$ for each credibility factor and each model–mechanism pairing.

CLI assignments were based on the pre-specified metrics and the empirical data. For each factor, we considered whether the evidence was specific to the mechanism, whether the models had been exercised under conditions commensurate with use, and whether the experimental comparators provided adequate resolution. The basis for each assigned CLI includes numeric comparisons or process evidence and is restated in the tables. We emphasize that we did not average scores across mechanisms or factors; instead, each is presented independently to clarify the contexts in which each model is suitable.

Numerical solver error — gradation.

- a. No quantification of discretization effects; reliance on default meshes
- b. Single refinement without focusing on regions impacting QoIs
- c. Targeted refinement in regions driving QoIs; solver tolerances explored
- d. Refinement, tolerance, and penalty studies demonstrate bounded QoI variation
- e. Error estimates are used to adjust acceptance bands or to widen decision margins

17. Results: factor-by-factor CLI ratings for each model×mechanism

Model H under M1 (walking) produced cortical strain fields that aligned with both gauge and DIC measurements at the superior and inferior neck. The normalized RMS error across all superior neck DIC points was 12.3% (gauges: 8.4%), and the LSI difference relative to tests was 0.03 absolute. The correlation between modeled and measured spatial strain patterns along the superior neck was $r = 0.84$. Mesh halving changed LSI by 0.9% and peak ϵ_1 by 1.2%, with solver tolerance tightening adding a further 0.2% change. The joint reaction vector was matched within 5.8% magnitude and 3.4° direction on average in the M1 test set.

Under M2 (stair descent with varus), Model H predicted a broader superior neck tensile band than observed in DIC. The peak ϵ_1 under varus was underpredicted by 22% on average across specimens, with the centroid offset by 6.1 mm proximally relative to DIC. The normalized RMS error across the superior neck DIC field was 18.7%. Mesh refinement altered peak ϵ_1 by 1.6% and centroid by 4.3 mm. The test rigs achieved varus

placement of $4.8 \pm 0.7^\circ$, joint reaction magnitude within 6.7%, and direction within 5.2° .

Model P under M1 produced normalized RMS errors of 7.4% (DIC) and 6.1% (gauges), with LSI difference 0.02 absolute relative to tests. The spatial correlation r between model and DIC strains was 0.89. Mesh refinement and solver tolerance changes altered LSI by 1.7% and peak ϵ_1 by 1.9%. The tests provided load reproduction fidelity as described above; perturbing the model loads by ± 1 SD around the measured values changed LSI by 1.5% on average.

Under M2, Model P captured the superior neck tensile band location and amplitude: the peak ϵ_1 error was 9.6% (model lower), with a centroid offset of 3.4 mm and a normalized RMS error across the DIC field of 11.2%. In sensitivity analyses, cohesive penalty variation changed peak ϵ_1 by 2.3% and band width by 0.9 mm. Mesh refinement changed peak ϵ_1 by 3.1%. Loads in the tests matched targets as above; perturbing model loads by ± 1 SD altered peak ϵ_1 by 4.2%.

Relevance of the validation activities to the COU — gradation.

- a. Validation data exist but with tenuous linkage to the COU loads and alignments
- b. Validation covers at least one load aligned with COU
- c. Validation maps closely to COU peaks; implant seating confirmed
- d. Validation includes alignment variation (varus) and imaging verification; load reproduction errors quantified
- e. Validation replicates both load vectors and muscle co-contractions with uncertainty bounds used in model perturbations

17.1. Results table: CLI ratings for Model H

The following table provides the per-factor CLI gradations for Model H under M1 (level walking) and M2 (varus stair descent). Each basis statement includes numeric evidence presented in the Results and Methods sections.

17.2. Results table: CLI ratings for Model P

The following table provides the per-factor CLI gradations for Model P under M1 (level walking) and M2 (varus stair descent). Each basis statement includes numeric evidence presented in the Results and Methods sections.

18. Discussion: implications for load transfer and stress shielding decisions within the COU

The two-model, two-mechanism assessment clarifies when a simplified representation suffices and when richer interfacial and material descriptions materially affect predicted risks. In the context of proximal load transfer during level walking, the bonded-interface, homogeneous bone idealization captured the

dominant redistribution of strain energy from the head–neck junction to the diaphysis, aligning satisfactorily with spatial strain patterns and with LSI differences that stayed within the predefined acceptance band. For the superior femoral neck under varus stair descent, by contrast, omitting interfacial separation smeared the tensile band and depressed the predicted peak, suppressing the risk signal that is clinically relevant for early fracture concerns.

Discretization effects and iterative tolerances were small relative to the acceptance margins across both models, with mesh refinement influencing the primary indices by less than 3.1% even in the more challenging cohesive/contact problems, and by less than 1.7% for the walking mechanism; these values are well below the bands set for strain and LSI agreement to measurements.

Agreement with measurements was consistent with the predefined bands in all but one pairing: the bonded-interface model for varus stair descent underpredicted peak tensile strain by 22% and misplaced the band centroid beyond the 5 mm tolerance. Where the patient-specific cohesive model was used, both the amplitude and the spatial placement of the tensile band fell within the bands, and the walking-case load share index and strain fields were close to the measured values in both models.

The chosen outputs inform the intended decisions directly: energy-based indices for walking reflected the proximal-to-diaphyseal partitioning that guides design trade-offs on cap stiffness and mantle thickness, and the tensile strain metrics for varus stair descent map to neck safety under the alignment scenario of interest.

The bench states reproduce the clinical activity peaks and alignment conditions used in the COU to a tolerable fidelity, with load magnitudes and directions aligned to instrumented-hip-informed values and with post-test imaging verifying varus placement and mantle characteristics; on this basis, the models were exercised under conditions that represent the clinical questions.

The applied loads and constraints matched their experimental counterparts within measured uncertainties that are small enough to not obscure the model-to-measurement differences reported, particularly in the walking case and in the varus stair descent case for the patient-specific cohesive model; the differences in the varus bonded-interface case exceed those uncertainties, underscoring the need for interfacial mechanics in that mechanism.

From a design and pre-clinical usage perspective, the findings support the following: for M1, both models can inform whether changes in component stiffness or mantle thickness would shift proximal load share beyond acceptable ranges, and the simpler model offers computational efficiency; for M2, assessing neck safety under varus stair descent requires the cohesive-interface, heterogeneous model to capture local debonding and strain localization. This mechanistic selectivity helps allocate modeling resources to align with where the design or alignment choices are sensitive to interfacial failure or bone heterogeneity.

Limitations include the focus on quasi-static peak loads, the absence of time-dependent behavior, and the constrained sample sizes. While these are appropriate for the present COU,

extension to fatigue or long-term remodeling decisions would require additional evidence, potentially including time-dependent mantle creep and density-driven adaptation models. The cadaveric setup reproduces muscle forces via surrogates that may differ from in vivo muscle wrapping; however, the load vector reconstructions and error bounds help anchor the relevance to the clinical activities. Finally, while the CT-derived orthotropy mapping follows established methods, individual errors in fabric direction could affect fine-scale strain fields; sensitivity analyses around these mappings would further strengthen confidence for extended uses.

Overall, the study demonstrates that a transparent, mechanism-focused credibility record enables both acceptance and non-acceptance determinations within a single COU, avoiding overgeneralization of a model beyond what the evidence supports.

Test conditions — gradation.

- a. No quantification of how well test loads and constraints match the modeled states
- b. Qualitative statement that tests are similar to modeled states
- c. Quantitative load matching reported; alignment assessed qualitatively
- d. Quantitative load and alignment matching reported with uncertainty; influence on QoIs discussed
- e. Quantitative matching with uncertainty propagated into model outputs; decision margins adjusted accordingly

19. Conclusions

We established the credibility of two implicit FEA models for hip resurfacing load transfer and stress shielding within a mechanism-focused COU using a private 0–5 Credibility Level Index. Under level walking (M1), both a homogeneous, bonded-interface model and a patient-specific, cohesive-interface model achieved agreement with measured strains and proximal load share within predefined acceptance bands and showed small discretization effects; consequently, either can be used to guide design choices that affect proximal load transfer in this activity.

Under varus stair descent (M2), only the patient-specific model with cohesive cement–bone and frictional cap–cement behavior captured the localized superior neck tensile band within amplitude and spatial tolerances. The bonded-interface model underpredicted the peak and diffused the band, indicating that it is not appropriate for judging neck safety in varus alignments under stair descent loads. Across both mechanisms, the selected quantities of interest align with the intended decisions, and the bench tests reproduce the essential loading and alignment conditions.

The resulting mechanism- and model-specific CLI grades, presented transparently with numeric bases, provide a usable

record under V&V40 principles. Future work will expand to time-dependent behaviors for remodeling and fatigue considerations and to additional alignment scenarios, using the present framework to extend credibility accordingly.

Output comparison — gradation.

- a.
Comparisons lack statistical treatment and predefined acceptance
- b.
Comparisons use some metrics but are interpreted without decision linkage
- c.
Comparisons include multiple metrics with bands and are referenced to the decision context
- d.
Comparisons include uncertainty propagation; mismatches analyzed for mechanism relevance
- e.
Comparisons and decisions are stress-tested across specimens and subpopulations with transparent criteria

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Table 1: Credibility Level Index (CLI) ratings for Model H by mechanism and factor

Credibility factor	Level	Basis
Model form - Model H - M1 (walking)	3.8	Homogeneous elastic bone with bonded interfaces reproduced proximal patterns (RMS strain 8.4% gauges, 12.3% DIC; LSI difference 0.03).
Numerical solver error - Model H - M1 (walking)	4.5	Mesh halving changed LSI by 0.9% and peak ϵ_1 by 1.2%; tighter tolerances changed QoIs by $\leq 0.2\%$.
Output comparison - Model H - M1 (walking)	3.9	Normalized RMS strain error 8.4% (gauges) and 12.3% (DIC); LSI mismatch 0.03 within 0.05 band.
Relevance of the quantities of interest - Model H - M1 (walking)	4.2	LSI and proximal cortical strains predicted remodeling stimulus; spatial correlation $r = 0.84$ supported interpretability.
Relevance of the validation activities to the COU - Model H - M1 (walking)	4.0	Mid-stance loads replicated within 5.8% magnitude and 3.4° direction; component seating confirmed by CBCT.
Test conditions - Model H - M1 (walking)	3.7	Applied joint and muscle forces matched targets within 5–11% and $3\text{--}4^\circ$; distal constraint mimicked physiologic support.
Model form - Model H - M2 (varus stair descent)	2.5	Bonded interfaces could not localize superior neck tensile band; peak ϵ_1 underpredicted by 22% and band broader than DIC.
Numerical solver error - Model H - M2 (varus stair descent)	4.4	Mesh halving changed peak ϵ_1 by 1.6% and centroid by 4.3 mm; tolerance sweeps affected QoIs by $< 0.4\%$.
Output comparison - Model H - M2 (varus stair descent)	3.0	Normalized RMS strain error 18.7%; peak ϵ_1 error 22%; centroid offset 6.1 mm, exceeding 5 mm band.
Relevance of the quantities of interest - Model H - M2 (varus stair descent)	4.0	Peak ϵ_1 and centroid along superior neck directly inform neck fracture risk under varus stair descent.
Relevance of the validation activities to the COU - Model H - M2 (varus stair descent)	3.6	Varus placement $4.8 \pm 0.7^\circ$ achieved; joint reaction matched within 6.7% and 5.2° for stair descent peak.
Test conditions - Model H - M2 (varus stair descent)	3.0	Load reproduction errors up to 9% magnitude and 6° direction; muscle surrogate errors up to 11%.

Table 2: Credibility Level Index (CLI) ratings for Model P by mechanism and factor

Credibility factor	Level	Basis
Model form - Model P - M1 (walking)	4.4	Orthotropic heterogeneity and cohesive/frictional interfaces produced RMS strain 6.1% (gauges), 7.4% (DIC), and LSI difference 0.02.
Numerical solver error - Model P - M1 (walking)	4.2	Refinement changed LSI by 1.7% and peak ϵ_1 by 1.9%; solver tolerance sweeps altered QoIs by <0.4%.
Output comparison - Model P - M1 (walking)	4.1	Normalized RMS strain error 6.1% (gauges), 7.4% (DIC); LSI mismatch 0.02 within 0.05 band; $r = 0.89$.
Relevance of the quantities of interest - Model P - M1 (walking)	4.5	Energy-based LSI and cortical strain fields map to remodeling stimulus with intact-normalized margins.
Relevance of the validation activities to the COU - Model P - M1 (walking)	4.2	Mid-stance loads replicated within 5.8% and 3.4°; specimen-specific imaging linked to model material maps.
Test conditions - Model P - M1 (walking)	3.9	Joint reaction and muscles reproduced within 5–10% and 3°; distal constraint verified; preconditioning stabilized fixtures.
Model form - Model P - M2 (varus stair descent)	4.3	Cohesive cement–bone and frictional cap–cement captured tensile band amplitude (9.6% error) and location (3.4 mm).
Numerical solver error - Model P - M2 (varus stair descent)	3.8	Refinement changed peak ϵ_1 by 3.1% and band centroid by 2.7 mm; penalty variation changed peak ϵ_1 by 2.3%.
Output comparison - Model P - M2 (varus stair descent)	4.0	Peak ϵ_1 error 9.6%; centroid offset 3.4 mm; normalized RMS error 11.2% across DIC field within 20% band.
Relevance of the quantities of interest - Model P - M2 (varus stair descent)	4.6	Superior neck peak tensile strain and band centroid are directly used to judge neck safety under varus descent loads.
Relevance of the validation activities to the COU - Model P - M2 (varus stair descent)	3.9	Varus placement ($4.8 \pm 0.7^\circ$) verified by CBCT; early-stance stair descent loads matched within 6.7% and 5.2°.
Test conditions - Model P - M2 (varus stair descent)	4.0	Loads and angles within 6–7% and 5°; muscle surrogates within 7–11%; alignment and mantle thickness verified.